

8/10/26

DATE:

PAGE:

Financial Functions

1 FV - future value

FV calculates how ~~no~~ much an investment or loan will be worth in the future after a certain period

Syntax: $=FV(\text{rate}, \text{nper}, \text{pmt}, [\text{pv}])$

rate = interest rate per period

nper = Number of periods

pmt = payment made each period

[pv] = present value / money ~~at~~ which is invested today

2 PMT - payment / EMI

PMT calculates the regular payment / EMI, required for a loan or investment

Syntax: $=PMT(\text{rate}, \text{nper}, \text{PV}, [\text{fv}])$

3 PPMT: Principal payment

Calculates a principal portion of a particular loan payment

when we pay EMI, the payment contains:
→ principal amt = The original amt of money

borrowed or invested, before adding interest

→ Interest amt = The extra money paid or earned on the principal amount

Ex: we have a loan of ₹50,000
monthly EMI is ₹2000

Suppose for the first month

→ PPMT = ₹1500 → reduces our loan from ₹50,000 to ₹48,500

→ IPMT = ₹500 → goes towards interest

Syntax: =PMT (rate, per, nper, pv)

per = it tells you which specific payment / period you want to calculate

Ex: • per = 1 → 1st month
• per = 5 → 5th month

Nper = How many period in total, tells you the total number of payments / periods.

Ex: • 3 year loan with monthly payments
• NPER = $3 \times 12 = 36$ months

per $\Rightarrow$ which one?

nper $\Rightarrow$ how much all together?

4 IPMT - Interest payment
calculates the interest portion of a particular loan payment
Syntax: $=IPMT(\text{rate}, \text{per}, \text{nper}, \text{pv})$

5 RATE - Interest rate
calculates the interest rate per period for a loan or investment
Syntax: $=RATE(\text{nper}, \text{pmt}, \text{PV}, [fv])$

6 PV - Present value
calculate the current value of the money based on a future amt, interest rate and time
Syntax: $=PV(\text{rate}, \text{nper}, \text{pmt}, [fv])$

7 NPER - Number of periods
calculates how long it will take to reach a particular financial goal or repay a loan
Syntax: $=NPER(\text{rate}, \text{pmt}, \text{pv}, [fv])$

period : one payment / investment interval,
such as month or year

RD - Recurring Deposit a fixed amount deposited
regularly, usually every month, to earn
interest

Questions

- 1 Mr Sunil purchased a hero honda karizma for Rs. 65000 by taking a loan from ICICI Bank, repayable in 3 years by monthly installments with interest @ 8.5% p.a. Calculate the EMI for the Loan [PMT]

$$= \text{PMT} (8.50\%/12, 3 \times 12, 65000)$$

$$= \text{PMT} (\text{rate}, \text{nper}, \text{PV}) \quad \text{o/p : 2051}$$

- 2 Mr Sunil wants to know the breakup of the principal amt in the EMI payable for the 1st month [PPMT]

$$= \text{PPMT} [\text{rate}, \text{per}, \text{nper}, \text{PV}, [\text{FV}]]$$

$$= \text{PPMT} (8.50/12, 1, 3 \times 12, 65000) \quad \text{o/p : 1591}$$

- 3 Now, Mr. Sunil wants to know the breakup of the interest amt in the EMI payable for the 1st month [IPMT]

$$= \text{IPMT}(\text{rate}, \text{per}, \text{nper}, \text{PV}[\text{FV}])$$

$$= \text{IPMT}(8.50\%/12, 1, 3 \times 12, 65,000) \quad \text{o/p: } 460$$

- 4 Ms. Meenakshi has a fixed deposit of Rs. 25000 in HDFC bank for 5 years on which interest given by the bank is 8.25%. Calculate the maturity value of Meenakshi [FV]

$$= \text{FV}(\text{rate}, \text{nper}, \text{pmt}, [\text{PV}])$$

$$= \text{FV}(8.25, 5, , 25000) \quad \text{o/p: } 37,160$$

- 5 Ms Rita is having the option of going for RA in which amount payable per month is Rs. 1200 interest @ 7.5% and the period of deposit will be 5 years, she asked you to calculate the maturity value for this option [FV]

$$= \text{FV}(\text{rate}, \text{nper}, \text{pmt}, [\text{PV}])$$

$$= \text{FV}(7.5/12, 5 \times 12, 1200) \quad \text{o/p: } 87,032$$

- 6 Mr. JP wants to invest in a plan where he will pay Rs. 1476 quarterly for 10 years to get a maturity value of Rs. 80,000. He wants you to calculate the rate of interest he will earn on this plan of investment [RATE]

$$= \text{RATE}(\text{nper}, \text{pmt}, \text{PV}, [\text{FV}])$$

$$= \text{RATE}(10 \times 4, , 1476, 80000) \quad \text{o/p : } 10.5\%$$

- 7 Before opting for the plan with the rate calculated above and for the same period & maturity value. Mr. JP now wants to know the present value of the investment plan [PV]

$$= \text{PV}(\text{rate}, \text{nper}, \text{pmt}, [\text{FV}])$$

$$= \text{PV}(10.50\%, 10 \times 4, , 80000) \quad \text{o/p : } -1476$$

- 8 Mrs. Bini wants to go for a RD scheme in any bank with an amt of ₹1520 per month where the interest is payable @ 7.5% per annum. She needs your help to find the no of installments she has to pay (also calculate the no. of years)

to get a maturity value of Rs 50000 [NPER

= NPER (rate, pmt, pv, [fv])

= NPER (7.5/12, 1520, ; 50000) o/p: 30